

Chaitanya Kasipally

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EXPERIENCE

Embedded Software Engineer II BWI Group

Sep2022 - Present(3+ Years)

Dayton, OH

- Working on developing and maintaining embedded software for VehicleDynamicsModule (VDM) through a combination of hand-coding (C) and auto-coding using ETAS RTACAR, DaVinci Developer, Configurator Pro & Matlab Simulink.
- Developed and validated damper control algorithms using MATLAB Simulink for Vehicle Dynamics applications; performed model-in-the-loop (MIL) and software-in-the-loop (SIL) testing and integrated auto-generated code into AUTOSAR-based ECU software.
- Led development and integration of the Diagnostic Communication Manager (DCM), DEM, and UDS service handling for Classic & Adaptive AUTOSAR ECUs on NXP/STM32 and Infineon Aurix platforms.
- Configured and validated Classic AUTOSAR BSW modules including OS, MCAL, IOHWAB, WDG, FEE, NvM, DCM, DEM, RTE.
- Analyze system-level and OEM requirements; perform requirement analysis, impact assessment, and raise formal change requests as per ASPICE-compliant processes.
- Debug production and development issues using Trace32, CANoe, UCAN, and internal testing tools; analyze memory/register states, diagnostic responses, and communication traces.
- Participate in cybersecurity software verification aligned with ISO 21434, including secure diagnostics and access control validation.
- Collaborate with cross-functional teams to define test strategies, automate regression testing, and support formal software releases. Ensure software quality and standard compliance using MISRA C, ISO 26262, and adhere to version control/review processes via Windchill and MKS.

Software Engineer Intern Motorola Solutions

May 2022 - August 2022 (3 Months)

Chicago, IL

- Worked on USB protocol stack testing for a proprietary embedded platform. Developed unit tests for control transfer logic, validated descriptor handling, and simulated USB enumeration behavior.

Software Engineer

August 2018 - August 2021 (3 Years)

Hyundai Motor India Engineering pvt ltd.

Hyderabad, India

- Developed embedded software and performed qualification, unit, interface, static, dynamic, and functional testing for Classic AUTOSAR SWCs and BSW modules, ensuring compliance with system and OEM requirements.
- Debugged complex functional issues across development and production ECUs, implementing corrective fixes and validating behavior through regression testing.
- Performed detailed timing analysis and model development on Infineon TC27x-based Powertrain Engine ECUs using TimingArchitect and Amalthea to evaluate task scheduling and execution constraints.
- Conducted Functional Safety testing in alignment with ISO 26262, validating Classic AUTOSAR components and Infineon-based ECU implementations.
- Executed qualification testing on Adaptive AUTOSAR software, with a focus on the Execution Manager running on NXP application processors in a Linux-based environment.
- Developed C++ test applications based on AUTOSAR Software Specification (SWS) documents to validate Adaptive BSW behavior on target hardware.
- Defined requirements and simulated real-time timing stress scenarios to analyze system behavior under load and propose timing optimization strategies. Designed and prototyped advanced pilot projects to evaluate new architectures, tooling, and performance improvements.
 - Real-time object detection for robotic platforms using *Convolutional Neural Networks (CNNs)*.
 - Sensor fusion algorithms for *lane assist* systems.
 - Automated temperature control systems for vehicle seating based on real-time sensor inputs.

SKILLS

Languages: Embedded C, C++, Assembly, Python, Java, .NET

AUTOSAR: Classic (SWC, RTE, OS, MCAL, DCM, DEM, FEE, WDG, NvM, IOHWAB), Adaptive (Execution Manager, Service Interfaces, SOME/IP)

Tools & IDEs: DaVinci Developer, DaVinci Configurator Pro, VTestStudio, Visual Studio, VSCode, MATLAB Simulink

Debugging & Analysis: Trace32, JTAG, CANoe, UCAN, Candela, TimingArchitect, Amalthea

Testing & Validation: LDRA, IBM RTRT (Unit & Interface Testing), Functional Testing, ISO 26262, ASPICE

Version Control & ALM: Git, Bitbucket, Windchill, PTC, JIRA

EDUCATION

Indiana Wesleyan University

Master's in Artificial Intelligence

Oct 2025 - Present

Relevant Coursework: Machine Learning, Computer Vision, Neural Networks, Optimization & Control Systems

University of Dayton

Master's in Computer Science | Certified in Autonomous Systems and Data Science | 3.9 GPA

Graduated: Dec 2022

Relevant Coursework: DataStructures, Algorithms, Artificial Intelligence, Deep Learning, Computer vision, Image Processing

ACHIEVEMENTS & AWARDS

Best Project - Won first prize in IEEE-WIE ENGINEERING PROJECT EXPO NNRG, competing 100+ projects

Dec 2017

Best Performer of the Quarter - For the project Timing Analysis on Powertrain Engine ECU

Sep 2019